



Urban Images Dataset

1,895 images
New classes: cyclist, e-scooter,
SUV, delivery van

Hybrid Autolabel Approach

YOLO-World for new classes
Manual validation & refinement
Quality assurance pipeline

Model Training

YOLOv8n architecture
9-class urban mobility
Optimized for edge deployment



Spatial Relationship Analysis

Person + Bicycle IoU >= 0.20
Geometric constraints validation
Unified cyclist class creation

Edge Optimization

ONNX to NCNN conversion
ARM CPU optimization
Memory efficiency tuning



Raspberry Pi 5 (8GB RAM)
Real-time processing
Privacy-preserving

9-class taxonomy
Real-time inference
Edge-optimized performance

KEY INNOVATIONS

ew urban mobility classes: cyclist, e-scooter, SUV, delivery van

dge deployment focus: Raspberry Pi 5 optimization with NCNN

brid autolabel approach: Semi-automated labeling methodology